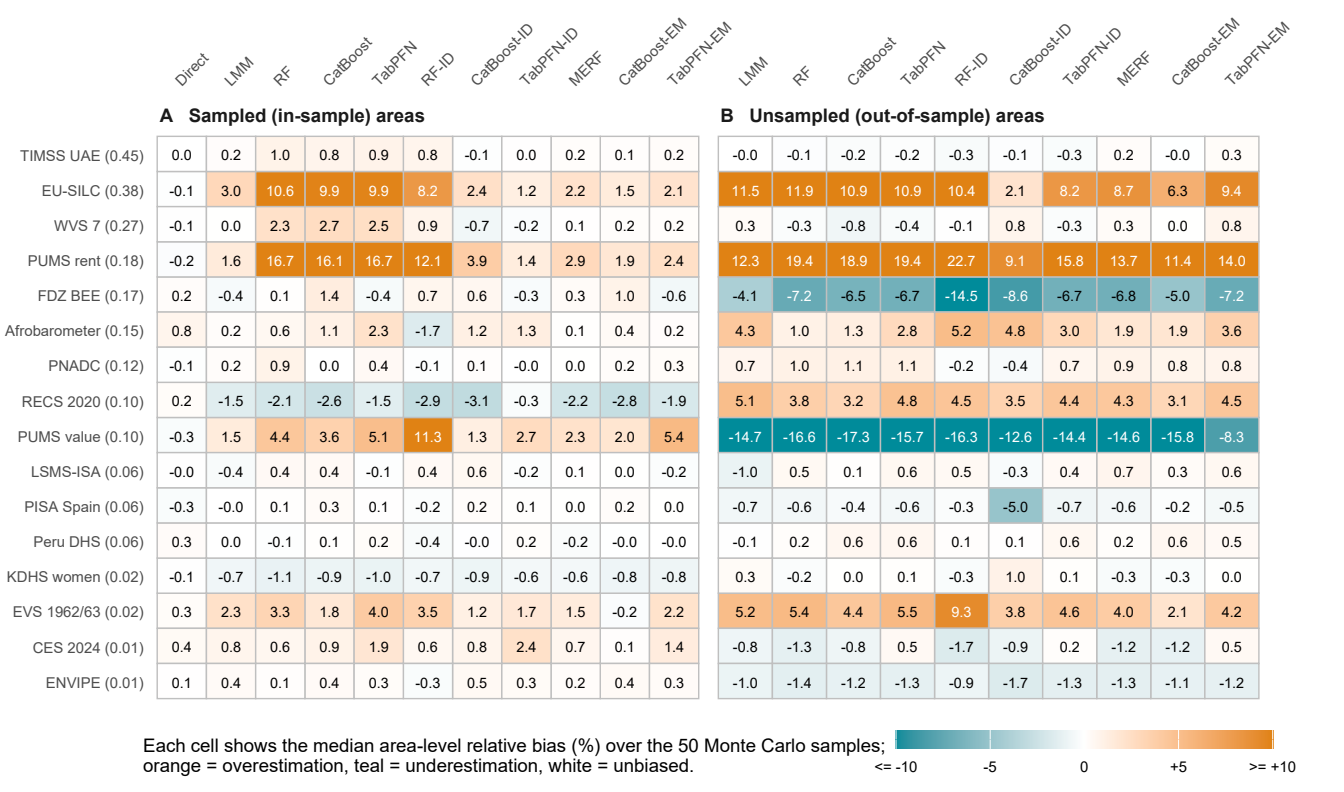
Direct
LMM
RF
CatBoost
TabPFN
RF-ID
CatBoost-ID
TabPFN-ID
MERF
CatBoost-EM
TabPFN-EM
A Sampled (in-sample) areas
TIMSS UAE (0.45) 0.0 0.2 1.0 0.8 0.9 0.8 -0.1 0.0 0.2 0.1 0.2
EU-SILC (0.38) -0.1 3.0 10.6 9.9 9.9 8.2 2.4 1.2 2.2 1.5 2.1
WVS 7 (0.27) -0.1 0.0 2.3 2.7 2.5 0.9 -0.7 -0.2 0.1 0.2 0.2
PUMS rent (0.18) -0.2 1.6 16.7 16.1 16.7 12.1 3.9 1.4 2.9 1.9 2.4
FDZ BEE (0.17) 0.2 -0.4 0.1 1.4 -0.4 0.7 0.6 -0.3 0.3 1.0 -0.6
Afrobarometer (0.15) 0.8 0.2 0.6 1.1 2.3 -1.7 1.2 1.3 0.1 0.4 0.2
PNADC (0.12) -0.1 0.2 0.9 0.0 0.4 -0.1 0.1 -0.0 0.0 0.2 0.3
RECS 2020 (0.10) 0.2 -1.5 -2.1 -2.6 -1.5 -2.9 -3.1 -0.3 -2.2 -2.8 -1.9
PUMS value (0.10) -0.3 1.5 4.4 3.6 5.1 11.3 1.3 2.7 2.3 2.0 5.4
LSMS-ISA (0.06) -0.0 -0.4 0.4 0.4 -0.1 0.4 0.6 -0.2 0.1 0.0 -0.2
PISA Spain (0.06) -0.3 -0.0 0.1 0.3 0.1 -0.2 0.2 0.1 0.0 0.2 0.0
Peru DHS (0.06) 0.3 0.0 -0.1 0.1 0.2 -0.4 -0.0 0.2 -0.2 -0.0 -0.0
KDHS women (0.02) -0.1 -0.7 -1.1 -0.9 -1.0 -0.7 -0.9 -0.6 -0.6 -0.8 -0.8
EVS 1962/63 (0.02) 0.3 2.3 3.3 1.8 4.0 3.5 1.2 1.7 1.5 -0.2 2.2
CES 2024 (0.01) 0.4 0.8 0.6 0.9 1.9 0.6 0.8 2.4 0.7 0.1 1.4
ENVIPE (0.01) 0.1 0.4 0.1 0.4 0.3 -0.3 0.5 0.3 0.2 0.4 0.3
LMM
RF
CatBoost
TabPFN
RF-ID
CatBoost-ID
TabPFN-ID
MERF
CatBoost-EM
TabPFN-EM
B Unsampled (out-of-sample) areas
-0.0 -0.1 -0.2 -0.2 -0.3 -0.1 -0.3 0.2 -0.0 0.3
11.5 11.9 10.9 10.9 10.4 2.1 8.2 8.7 6.3 9.4
0.3 -0.3 -0.8 -0.4 -0.1 0.8 -0.3 0.3 0.0 0.8
12.3 19.4 18.9 19.4 22.7 9.1 15.8 13.7 11.4 14.0
-4.1 -7.2 -6.5 -6.7 -14.5 -8.6 -6.7 -6.8 -5.0 -7.2
4.3 1.0 1.3 2.8 5.2 4.8 3.0 1.9 1.9 3.6
0.7 1.0 1.1 1.1 -0.2 -0.4 0.7 0.9 0.8 0.8
5.1 3.8 3.2 4.8 4.5 3.5 4.4 4.3 3.1 4.5
-14.7 -16.6 -17.3 -15.7 -16.3 -12.6 -14.4 -14.6 -15.8 -8.3
-1.0 0.5 0.1 0.6 0.5 -0.3 0.4 0.7 0.3 0.6
-0.7 -0.6 -0.4 -0.6 -0.3 -5.0 -0.7 -0.6 -0.2 -0.5
-0.1 0.2 0.6 0.6 0.1 0.1 0.6 0.2 0.6 0.5
0.3 -0.2 0.0 0.1 -0.3 1.0 0.1 -0.3 -0.3 0.0
5.2 5.4 4.4 5.5 9.3 3.8 4.6 4.0 2.1 4.2
-0.8 -1.3 -0.8 0.5 -1.7 -0.9 0.2 -1.2 -1.2 0.5
-1.0 -1.4 -1.2 -1.3 -0.9 -1.7 -1.3 -1.3 -1.1 -1.2
Each cell shows the median area-level relative bias (%) over the 50 Monte Carlo samples;
orange = overestimation, teal = underestimation, white = unbiased.
<= -10
-5
0
+5
>= +10